

Bayesian Black-Box Optimisation

Final research report (evidence pack)

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https://github.com/Neuroxa-Labs/Bayesian_Optimisation

Status: Weeks 1-12 complete - Week 13 queries locked - portal y pending

Primary method: GP (Matern + ARD) - EI / UCB - trust-region exploit

Report version: v1 (pre-Week-13 results)

Imperial College London - PCMLAI Stage 2 - Neuroxa-Labs

1. Objective

Maximise eight unknown continuous black-box functions $f_i : [0,1]^{d_i} \rightarrow \mathbb{R}$ under a strict budget of one query per function per week. True formulas are withheld; only portal evaluations (x, y) are observed. Higher y is always better. The goal is not to recover the formula; it is to find high-performing x with sample-efficient sequential decisions.

2. Overall summary (after Week 12)

- F1 Radiation 2D | best 7.711e-16 | seed incumbent; late signal lobe unresolved
 - F2 Noisy ML 2D | best 0.776645 | strong sharp ridge
 - F3 Drug/adverse 3D | best -0.011366 | safe local band (x3 lock)
 - F4 Warehouse 4D | best 0.678600 | strong late climb (Week 12)
 - F5 Chem. yield 4D | best 3800.74 | ridge optimum (x1 climb)
 - F6 Cake 5D | best -0.136 | Week-10 basin; fragile
 - F7 HP tuning 6D | best 1.872233 | strong late climb (Week 12)
 - F8 8-param ML 8D | best 9.872928 | strong late climb (Week 12)
- Late improve counts: W8 3/8 - W9 4/8 - W10 5/8 - W11 4/8 - W12 4/8 (F4, F5, F7, F8).

Best-so-far y · Seed and Weeks 8-12 (bold = improved vs previous column)

Fn	Task	D	Seed	W8	W9	W10	W11	W12
F1	Radiation	2	7.711e-16	7.711e-16	7.711e-16	7.711e-16	7.711e-16	7.711e-16
F2	Noisy ML	2	0.6112	0.7766	0.7766	0.7766	0.7766	0.7766
F3	Drug / adverse	3	-0.0348	-0.0114	-0.0114	-0.0114	-0.0114	-0.0114
F4	Warehouse	4	-4.026	0.5720	0.6420	0.6670	0.6752	0.6786
F5	Chem. yield	4	1089	3760	3769	3779	3790	3801
F6	Cake recipe	5	-0.7143	-0.2404	-0.2404	-0.1360	-0.1360	-0.1360
F7	HP tuning	6	1.365	1.857	1.858	1.863	1.866	1.872
F8	8-param ML	8	9.598	9.868	9.869	9.871	9.872	9.873

Figure: Compact incumbent / best-so-far by week.

Trends influencing next query — best so far

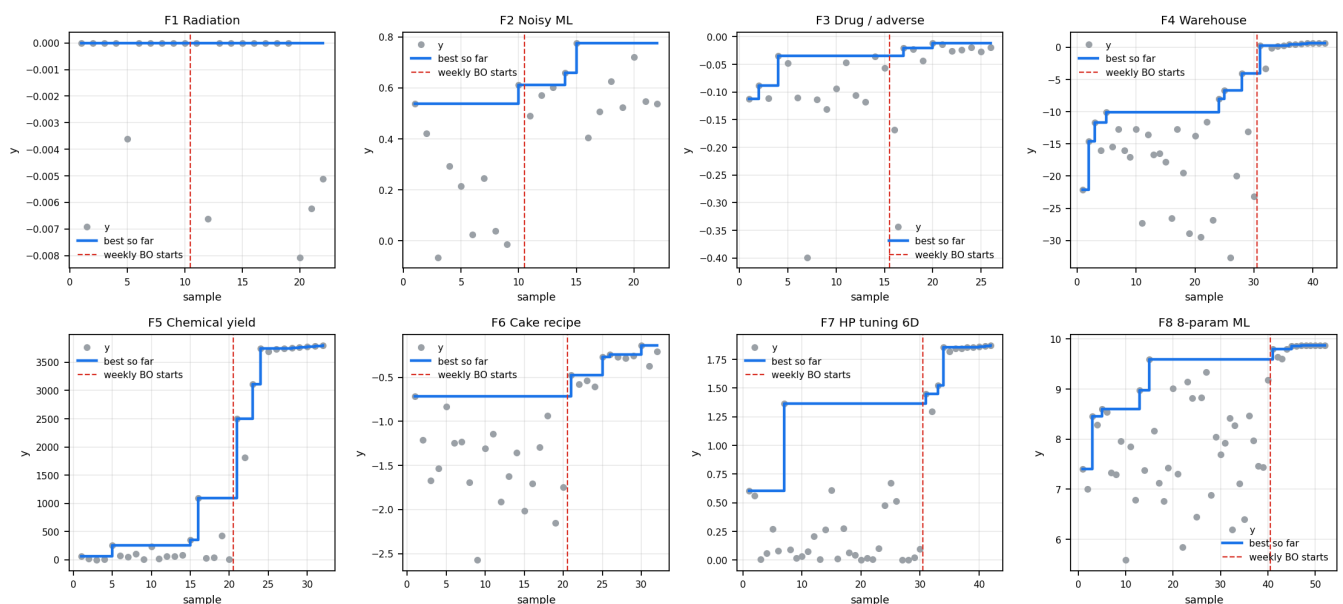


Figure: Best-so-far curves across functions.

3. Method

3.1 Core loop

- Load per-function history from data/function_/initial_.npy.
- Fit a Matern GP with ARD length scales (multi-restart LML).
- Score candidates with EI or UCB; search globally and inside a trust region.
- Apply per-function constraints (locks, boundary penalty, anti-duplicate, F1 trust gate).
- Submit a six-decimal portal string; append y; repeat.

Executable code: notebooks/BBO_Capstone_Optimized.ipynb. Transparency: DATASHEET.md - MODEL_CARD.md. Technical rationale: docs/TECHNICAL_JUSTIFICATION.md.

3.2 Per-function specialisations

- F1: trust gate - refuse GP exploit while labels are ~null; exploit after signal cluster.
- F2: WhiteKernel noise model; hard-return to sharp ridge after misses.
- F3: safe x3 lock near the -0.011 neighbourhood.
- F4 / F7 / F8: shrinking trust-region micro-steps on ARD-sensitive axes.
- F5: log-y GP fit; lock high x2-x4 face; climb x1.
- F6: hard-return to Week-10 cake centroid after failed neighbour steps.

3.3 What we deliberately did not do

Neural nets / large ensembles were not the primary weekly decider. With $n \sim 20$ -50 points, calibrated uncertainty matters more than flexible curve-fitting. Peer pipelines that use NN/Optuna/TuRBO as diagnostics are compatible with this view.

4. Winning-query evidence

- F1 7.711e-16 - Seed (late lobe still below seed max)
- F2 0.776645 - Week 5 GP/EI on noisy ridge
- F3 -0.011366 - Week 6 local exploit + x3 discipline
- F4 0.678600 - Week 12 trust-region micro-step
- F5 3800.74 - Week 12 ridge climb ($x_1=0.44$)
- F6 -0.136 - Week 10 local basin hit
- F7 1.872233 - Week 12 trust-region micro-step
- F8 9.872928 - Week 12 trust-region micro-step

Takeaway: different functions rewarded different regimes - early ridge discovery, mid-project locks, and late compressed exploit. F1 remains sparse detection; F6 shows leaving a sharp basin without evidence is expensive.

5. Per-function findings

F1 - Radiation source (2D)

Locate a hidden source; most of the map reads ~ 0 . Incumbent $y=7.711\text{e-}16$ at $[0.731024, 0.733000]$ (seed). Weeks 10-12 opened a measurable lobe near $(0.64, 0.68)$ with readings $-0.00807 \rightarrow -0.00623 \rightarrow -0.00512$ (still below seed max). Unresolved absolute peak; final round stays in the signal lobe.

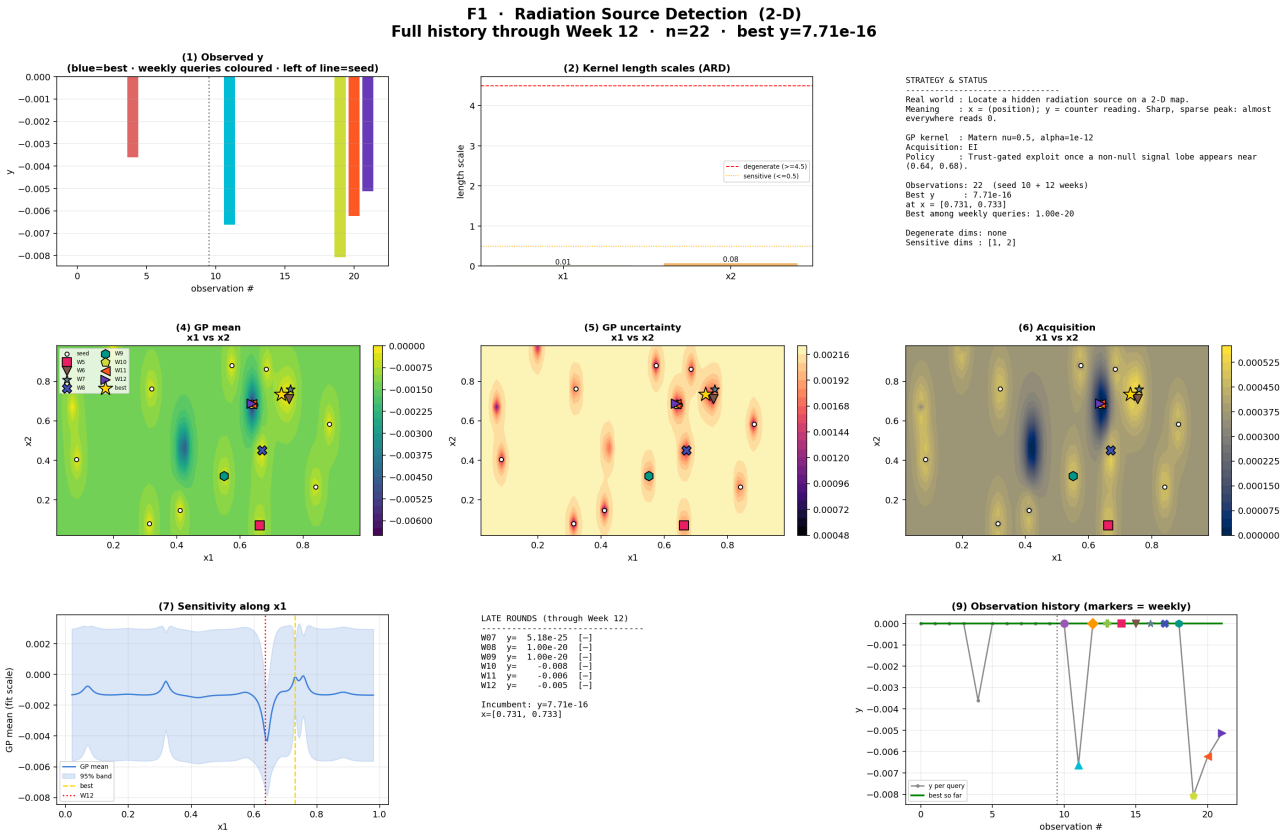


Figure: F1 - Radiation source (2D)

F2 - Noisy ML score (2D)

Incumbent $y=0.776645$ at $[0.717869, 0.020000]$. Sharp ridge: nearby steps often land $\sim 0.54 \rightarrow$ hard-return policy.

F2 · Noisy ML Log-Likelihood (2-D)

Full history through Week 12 · n=22 · best y=0.777

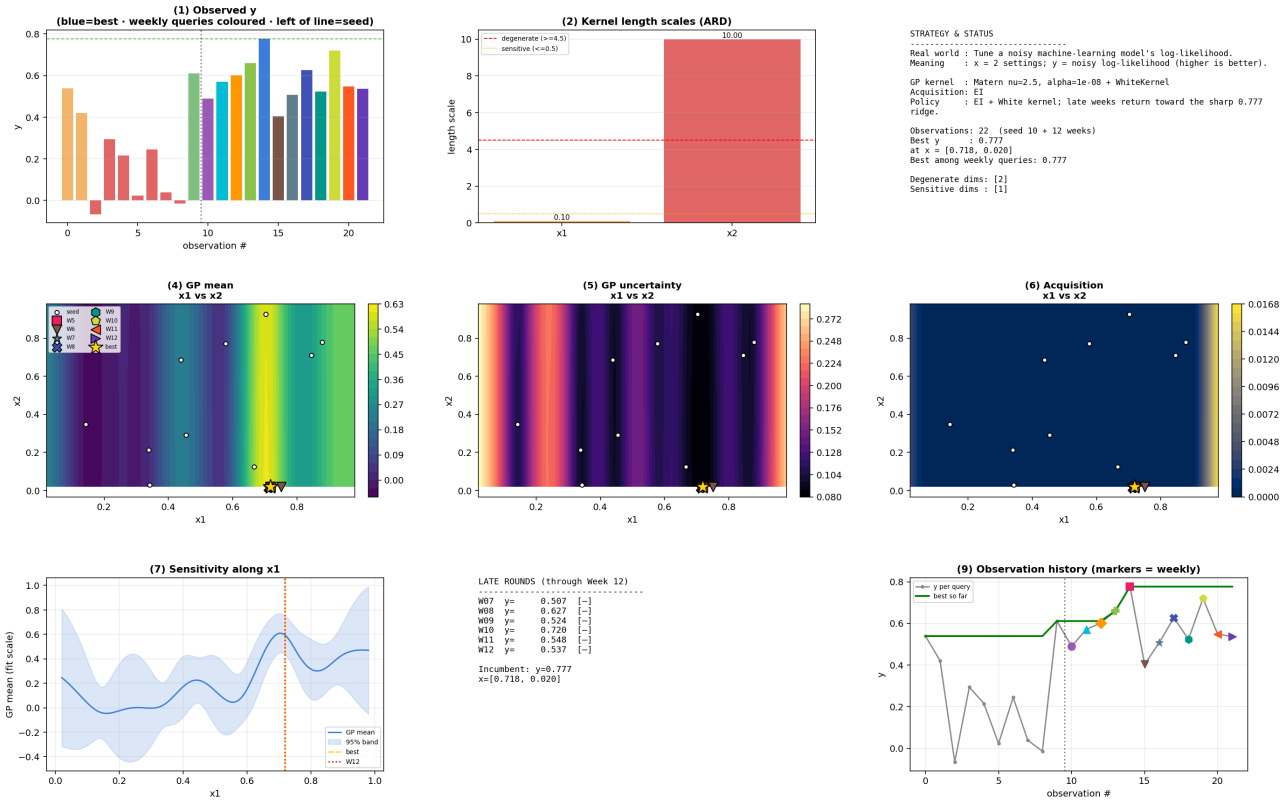


Figure: F2 - Noisy ML score (2D)

F3 - Drug / adverse (3D)

Incumbent y=-0.011366 at [0.492581, 0.691593, 0.401000]. x3 is sensitive; lock the safe band. History has 11 weekly points in data/.

F3 · Drug Discovery - Adverse Reactions (3-D)

Full history through Week 11 · n=26 · best y=-0.011

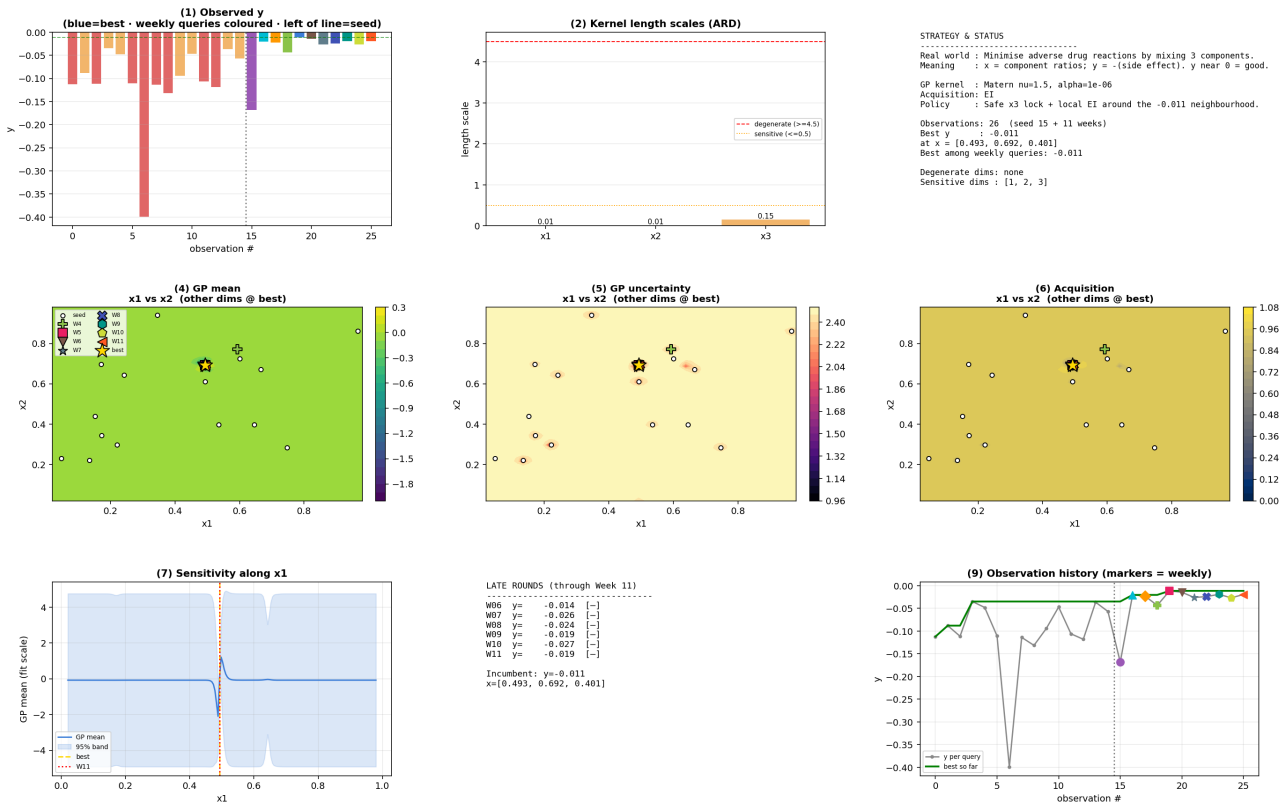


Figure: F3 - Drug / adverse (3D)

F4 - Warehouse (4D)

Incumbent $y=0.678600$ (Week 12). After a basin is proven, micro-steps beat global jumps.

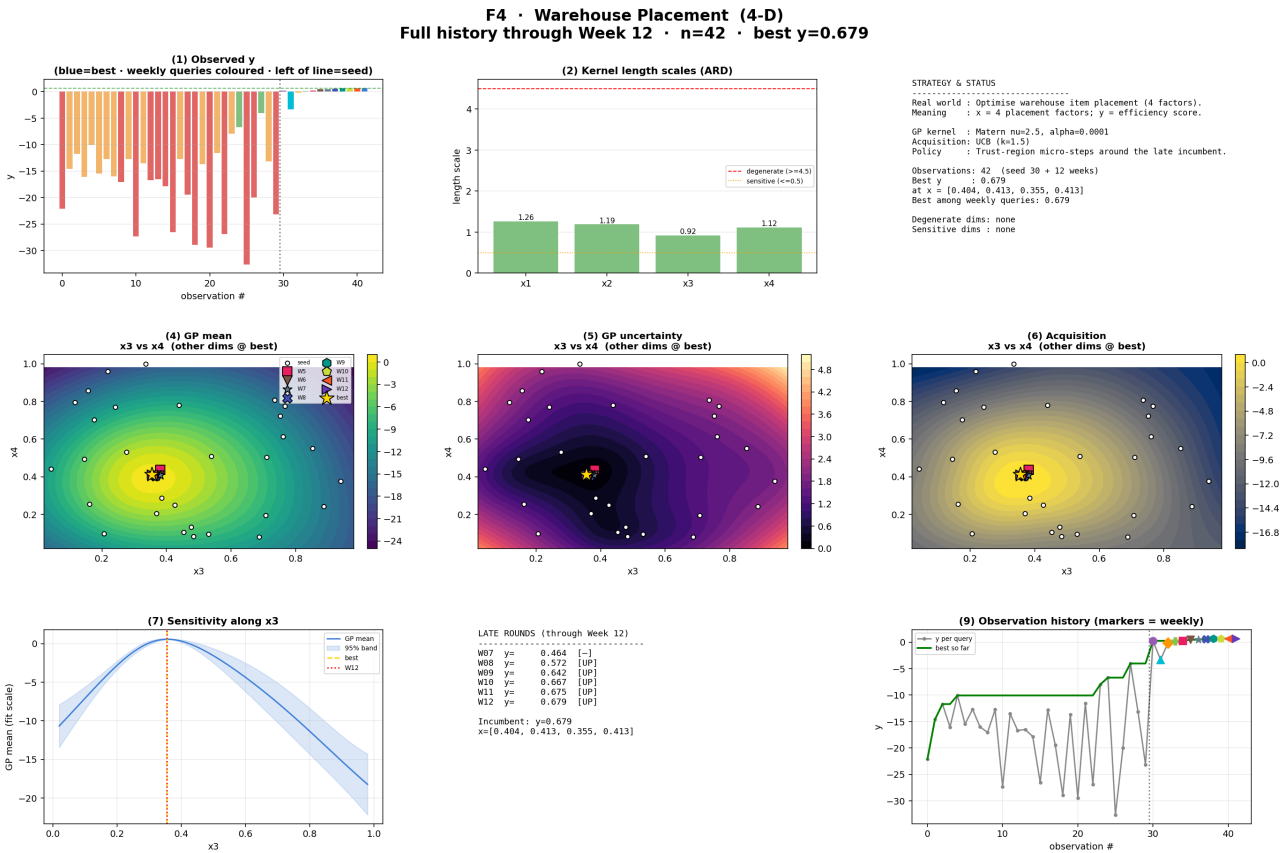


Figure: F4 - Warehouse (4D)

F5 - Chemical yield (4D)

Incumbent $y=3800.74$ at $[0.44, 0.98, 0.98, 0.98]$. Clearest success: lock high face, climb x_1 (seed ~1089 -> ~3801).

F5 • Chemical Yield Optimisation (4-D)

Full history through Week 12 • n=32 • best y=3800.740

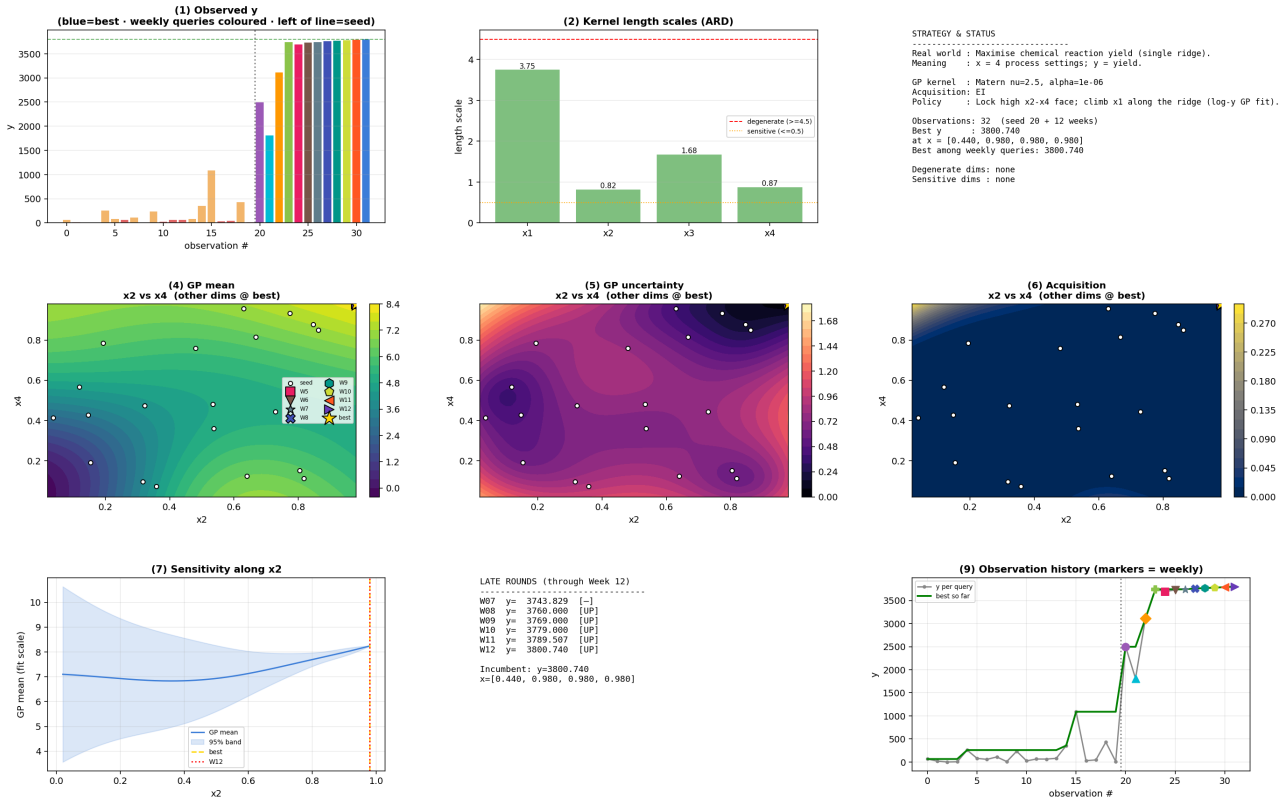


Figure: F5 - Chemical yield (4D)

F6 - Cake recipe (5D)

Incumbent $y=-0.136$ (Week 10). Week 11 collapse after a small off-centroid step → hard-return rule.

F6 • Cake Recipe Optimisation (5-D)

Full history through Week 12 • n=32 • best $y=-0.136$

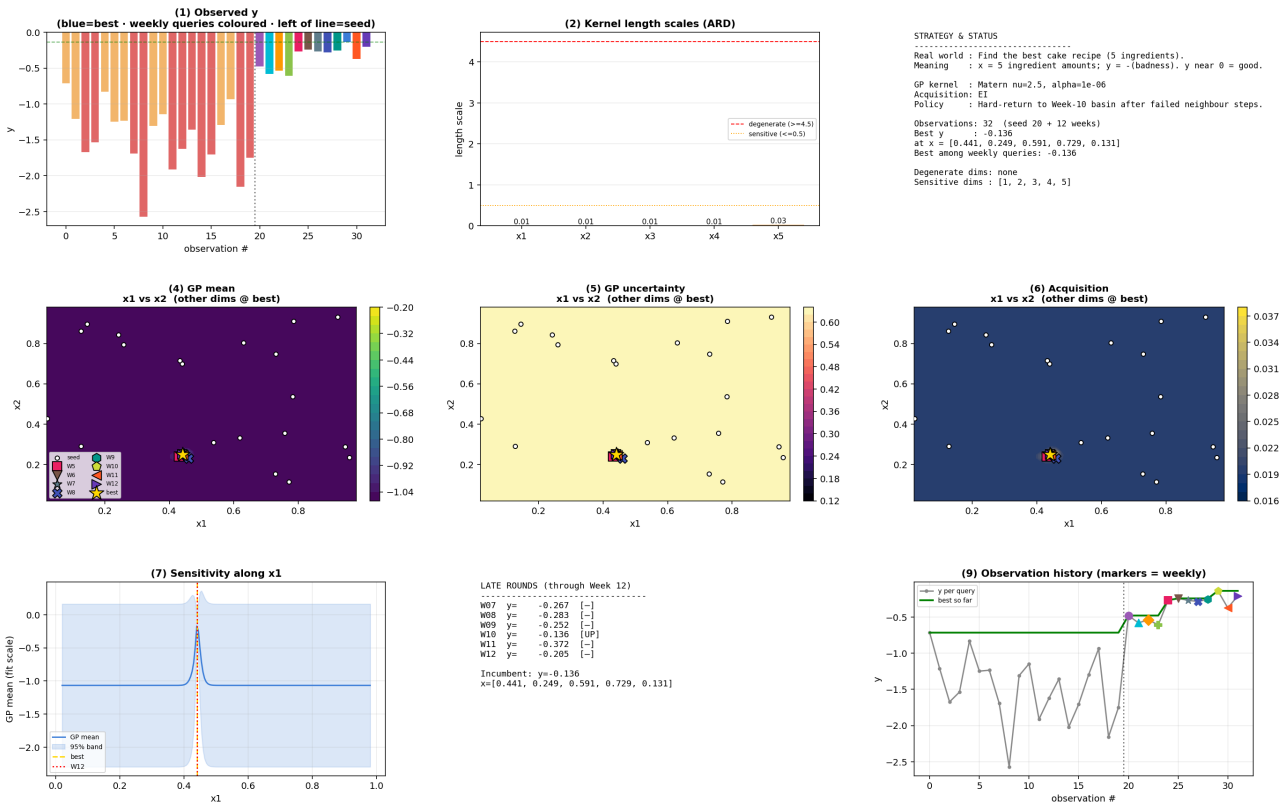


Figure: F6 - Cake recipe (5D)

F7 - Hyperparameter tuning (6D)

Incumbent $y=1.872233$ (Week 12). Move ARD-sensitive axes only; accept slow compound gains.

F7 · ML Hyperparameter Tuning (6-D) Full history through Week 12 · $n=42$ · best $y=1.872$

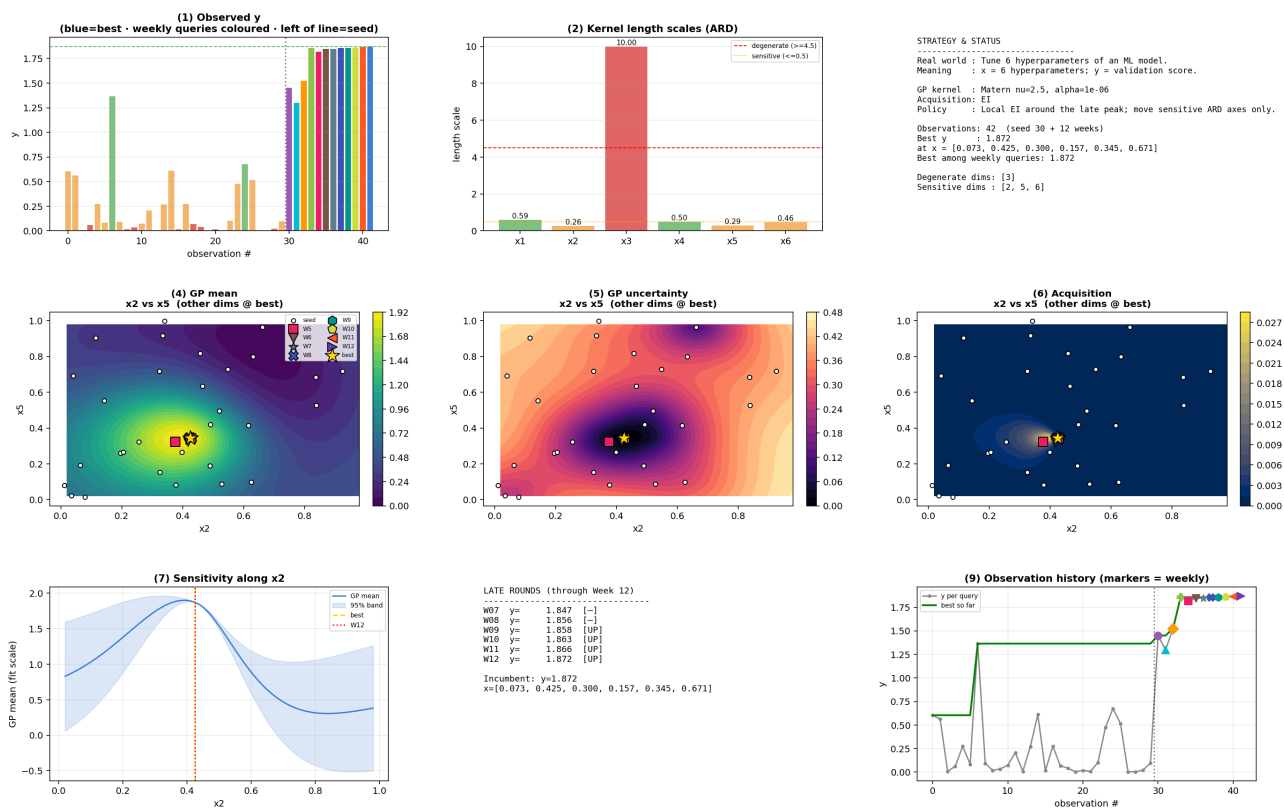


Figure: F7 - Hyperparameter tuning (6D)

F8 - Eight-parameter ML (8D)

Incumbent $y=9.872928$ (Week 12). Trust-region ticks + boundary penalty; do not chase edge sigma artefacts.

F8 · 8-Parameter ML Model Optimisation (8-D)
Full history through Week 12 · n=52 · best y=9.873

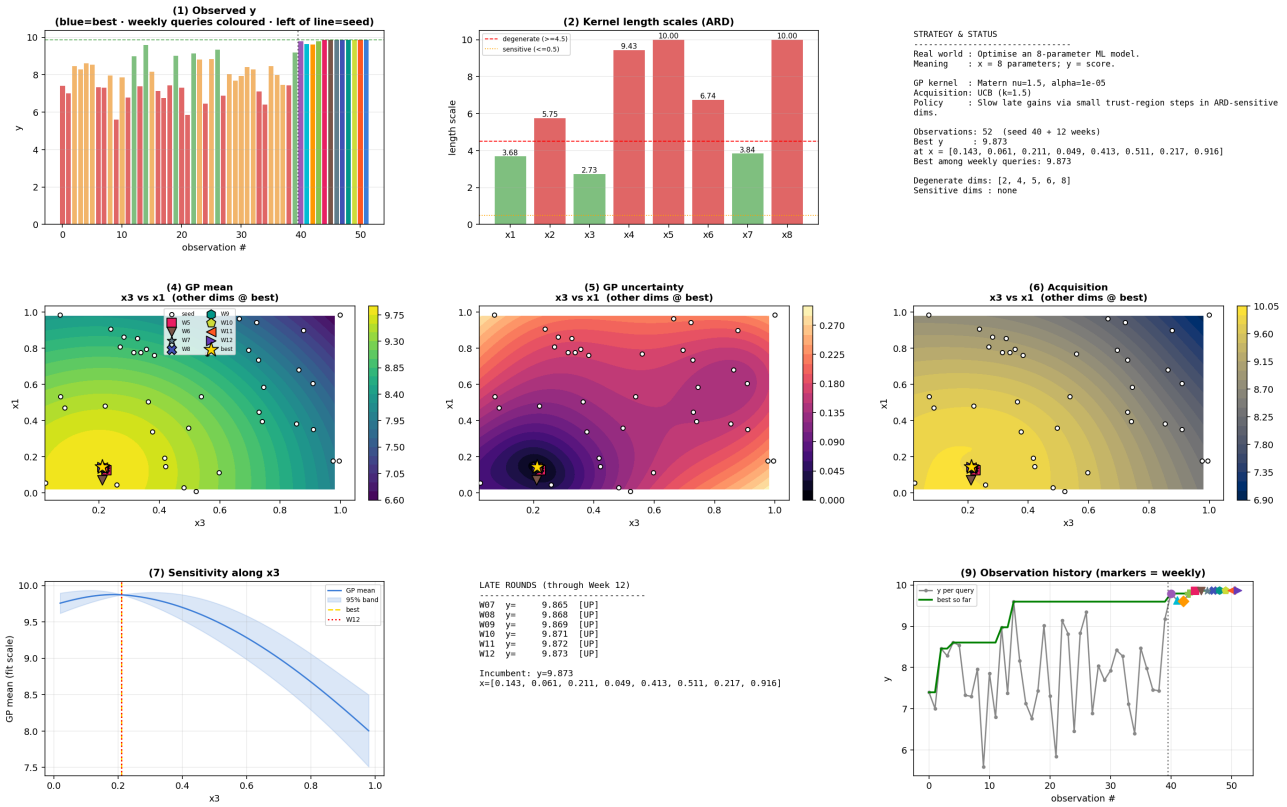


Figure: F8 - Eight-parameter ML (8D)

Cluster view (Week 12)

Promising clusters (KMeans hulls) · axes = ARD-sensitive dims · ★ = best y

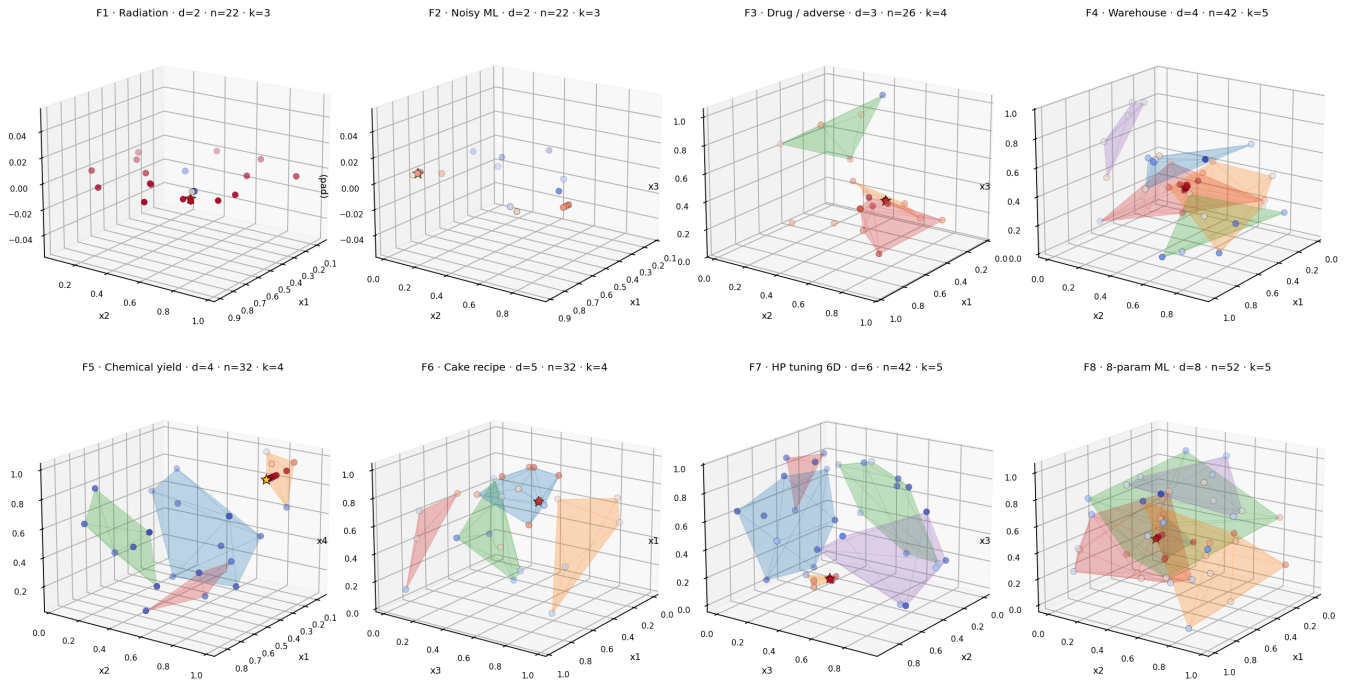


Figure: 3D cluster gallery across functions.

6. Week 13 / final round (queries locked)

Near-pure exploitation and recoveries. Full rationale: weeks/WEEK13_STRATEGY.md. RL framing: weeks/final_round_rl_reflection.md.

```
F1: 0.635000-0.688000
F2: 0.717870-0.020000
F3: 0.492580-0.691590-0.401000
F4: 0.405000-0.412000-0.354000-0.414000
F5: 0.450000-0.980000-0.980000-0.980000
F6: 0.441200-0.249200-0.590800-0.728700-0.131200
F7: 0.074000-0.424000-0.299000-0.158000-0.346000-0.672000
F8: 0.144000-0.060000-0.210000-0.050000-0.414000-0.510000-0.216000-0.917000
```

- F1: signal-lobe micro-step
- F2 / F3 / F6: hard return toward historical best
- F4 / F7 / F8: micro from Week-12 incumbents
- F5: ridge continue x1=0.45

Portal y for Week 13: pending - summary table will be updated when results arrive (scripts/append_week13.py).

7. Limitations

- One query per function per week - no inner real-function line search.
- GP can be confidently wrong in empty regions (inflated sigma -> boundary chase).
- Clustered sampling can miss a distant second mode.
- F1 absolute best remains a near-null seed reading despite a late measurable lobe.
- F2 observed best may be an optimistic draw under noise.
- F3 weekly count in data/ is 11 (not 12); no fabricated point was added.

8. Related artefacts

- Approach presentation text - docs/approach_presentation.md
- Project reflection - weeks/project_reflection.md
- Successful strategies (+ Matt peer) - weeks/successful_strategies_reflection.md
- Visual gallery - reports/analysis/README.md
- Progress dashboard - reports/progress/README.md
- Course file map - docs/COURSE_INDEX.md
- Project FAQ (PDF) - docs/project_faq.pdf

9. Conclusion

Under a one-query-per-week budget, a single interpretable GP per function, specialised with ARD, locks, trust regions and an F1 trust gate, produced clear late gains on F4/F5/F7/F8, a strong ridge on F2, a safe band on F3, and a fragile but real basin on F6. F1 remains the hardest sparse-signal case. Week 13 commits to near-pure exploitation of those validated regions. When portal results return, the overall table and winning-query section of this report will be revised in place - without changing the method narrative.